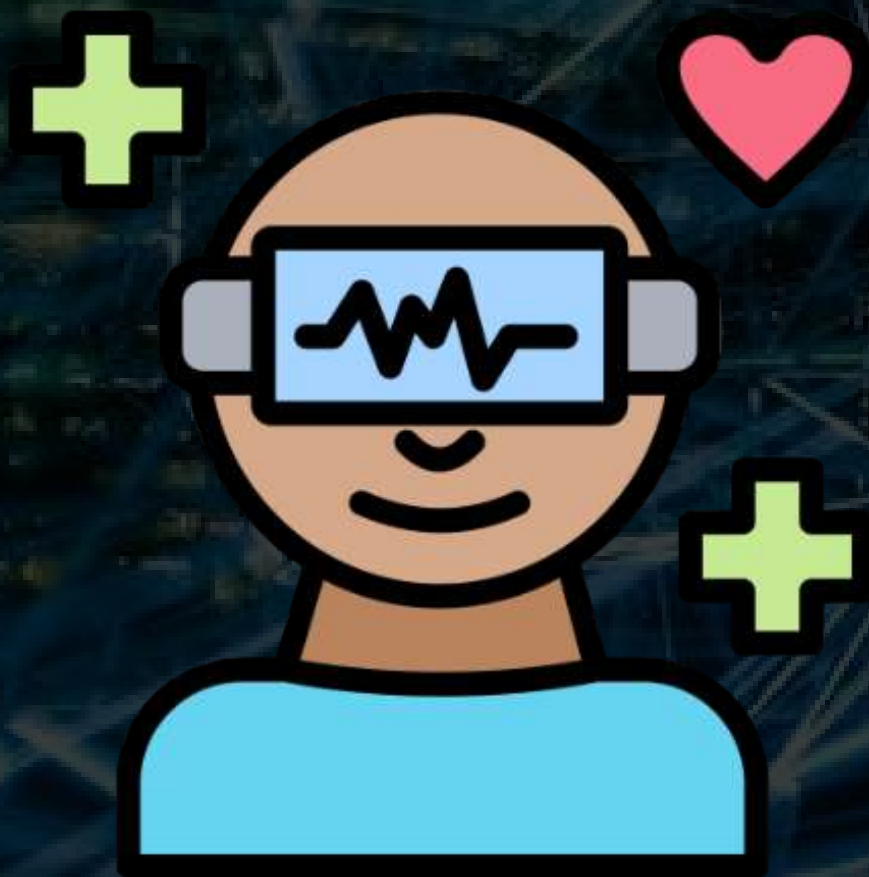


SOFTWARE DEVELOPMENT APPLIED TO INTERCULTURALITY

Embodying Affect: A Review of Emotion Modeling, Detection, and Expression in Immersive Virtual Agents

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WHY AFFECTIVE COMPUTING IN VR?



- **The Role of Emotion:** Essential for rational decision-making, perception, and learning.
- **VR Potential:** Powerful tool for simulating complex real-world situations in controlled conditions.
- **The Goal:** “Interactive Virtual Reality Environments” that interpret and react to human emotions.

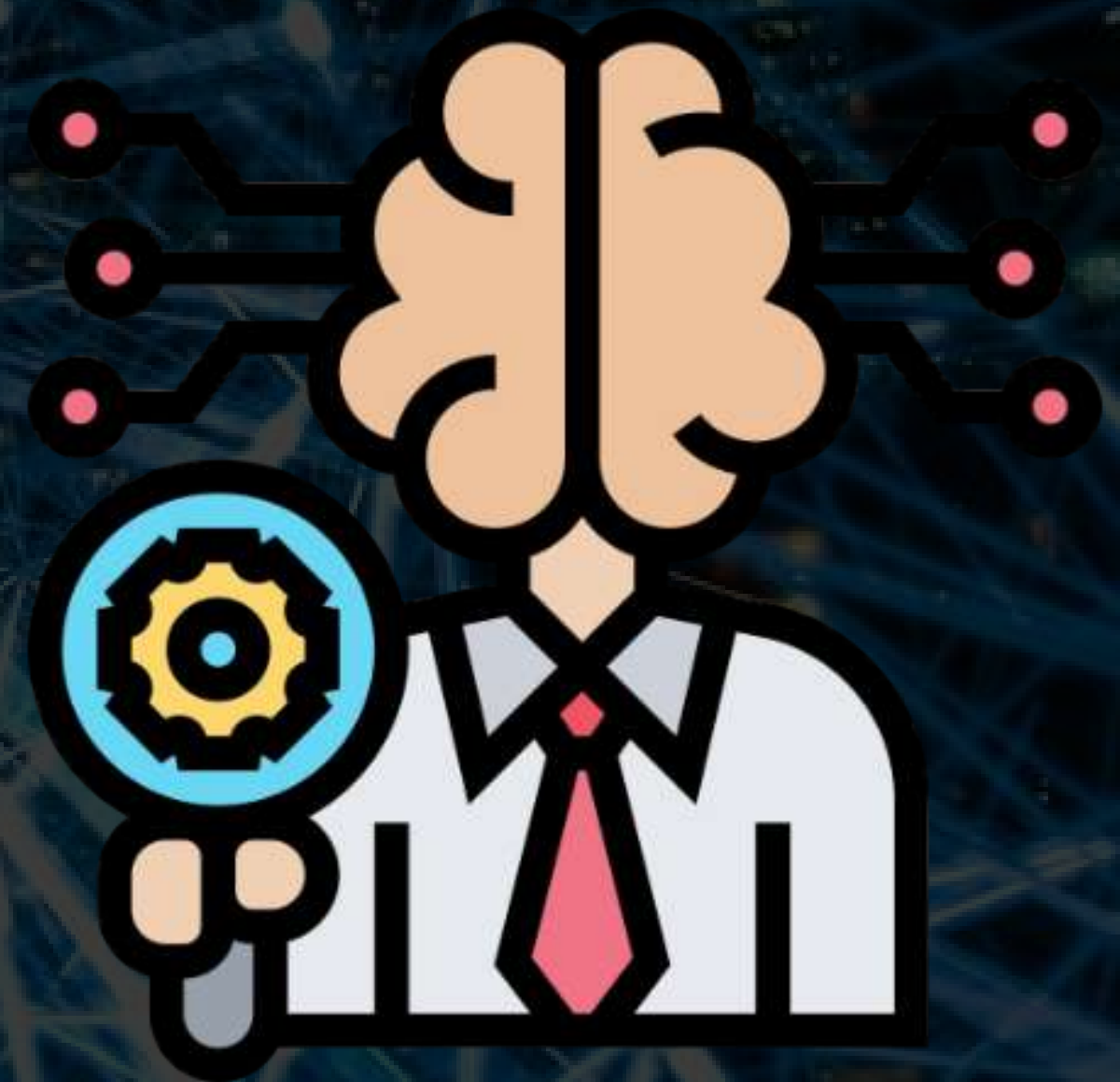
COMPUTATIONAL MODELS OF EMOTION

- **Appraisal Framework**

- Connects emotion with cognition.
- Emotions arise from evaluating events based on beliefs, desires, and intentions.
- Common models: OCC (Ortony–Clore–Collins) and Lazarus.

- **Dimensional Framework**

- Represents emotions on continuous axes.
- Most used: Circumplex Model of Affect (CMA) with Valence, Arousal, and Dominance.
- Useful for affect recognition and continuous emotional representation.



MULTIMODAL EMOTION DETECTOR

Why Multimodal?

Emotion-aware systems need to detect what the user feels without breaking immersion. Self-reports interrupt the experience, so current research focuses on physiological signals that operate silently in the background.

Physiological Signals and Deep Learning

Physiological cues like heart-rate variability and skin conductance offer implicit emotional data, but individually they lack the precision needed for complex states. Deep learning models that fuse multiple signals—such as eye-tracking with galvanic skin response—have shown far superior accuracy.



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MULTIMODAL EMOTION DETECTOR

Multi-Domain Sensing

Capturing signals from different anatomical regions strengthens emotional inference even further. Models that merge head-based eye data, trunk ECG, and peripheral EDA provide a more complete picture of the user's arousal and valence, consistently outperforming single-domain approaches.



EXPRESSIONS AND INTERACTION

Avatar Expressiveness and Realism

Avatars enhance social presence, but higher facial expressiveness doesn't necessarily change the user's emotional state. What matters most is the synchrony between voice and gesture; users quickly notice mismatches. Even a mix of natural and synthetic cues can create strong empathy if at least one modality is high-fidelity.



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EXPRESSIONS AND INTERACTION

Affective Visualization



VR can display the user's internal state as visual feedback. Representing stress or relaxation within the environment creates a feedback loop that supports emotional self-regulation and therapeutic use.

CONCLUSION: FUTURE DIRECTIONS



Current research in VR is shifting toward closed-loop affective systems, where emotions are continuously detected, modeled, and expressed. Detection relies on deep learning that fuses multiple physiological signals, modeling depends on interoperable cognitive frameworks, and expression blends natural and synthetic cues to sustain empathy. The next challenge is testing these systems in richer, more interactive environments to bring them closer to real-world use.



THANK YOU!